

Week 4 Password Security Practice

Part 1: MD5 Hash Cracking Results

Five MD5 hashes were generated using md5.cz and an attempt was made to crack them using CrackStation (crackstation.net).

Password	MD5 Hash	Result
password123	482c811da5d5b4bc6d497ffa98491e38	Cracked
genius*12\$	b517349ad50faf3d0c48aaa6cc93a51b	Not Cracked
see_you%	3ad51b52ca10858247a8c7c2dacb98a0	Not Cracked
rea_L&CHar109	3d10abd4f5c6c09ea071888d1c8dbbef	Not Cracked
Pret(34#sh	12639f840f3fa18e476f12d119ffbc28	Not Cracked

Observation

- Only "password123" was cracked because it is a common, predictable password that already exists in CrackStation's pre-built dictionary of known/leaked passwords.
- The remaining four passwords were not cracked because they are unique, random, and use a mix of letters, numbers, and special characters. They do not match any word in the dictionary, so the tool could not reverse the hash.
- This confirms that hash-cracking tools work by dictionary/pattern matching, not by mathematically reversing the hash. A strong, unpredictable password is what keeps a hash safe from being cracked.

Part 2: Zynxis Password Policy

To protect company accounts and data, all Zynxis employees and interns must follow the password rules below.

1. Minimum Length

- Passwords must be at least 12 characters long.
- Longer passwords are harder to crack, even with automated tools.

2. Character Requirements

- Must include at least one uppercase letter (A-Z).
- Must include at least one lowercase letter (a-z).
- Must include at least one number (0-9).
- Must include at least one special character (e.g. ! @ # \$ % & *).

3. How Often to Change

- Passwords should be changed every 90 days (3 months).
- Passwords must be changed immediately if a data breach or suspicious login activity is suspected.

4. What Not to Use

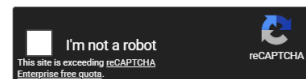
- Do not use your name, username, or company name.
- Do not use your birthdate, phone number, or other personal information.
- Do not use common passwords (e.g. password123, 123456, qwerty, admin).
- Do not reuse the same password across multiple accounts or platforms.
- Do not use simple keyboard patterns (e.g. asdfgh, 111111).

5. Additional Recommendations

- Use a unique password for every account.
- Enable Two-Factor Authentication (2FA) wherever available.
- Use a password manager to generate and store strong passwords securely.
- Never share your password with anyone, including coworkers or IT staff.

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
482c811da5d5b4bc6d497ffa98491e38	md5	password123

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

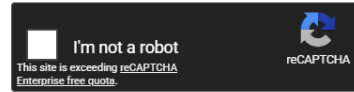
[Download CrackStation's Wordlist](#)

How CrackStation Works

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

b517349ad50faf3d0c48aaa6cc93a51b



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Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
b517349ad50faf3d0c48aaa6cc93a51b	Unknown	Not found.

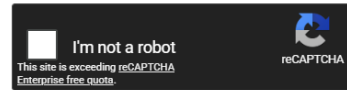
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3ad51b52ca10858247a8c7c2dacb98a0



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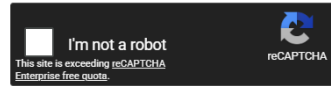
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3d10abd4f5c6c09ea071888d1c8dbbef



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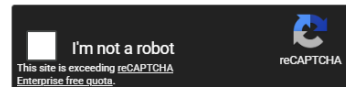
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